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Proposal for:	Proposal Date:	Proposal Number:
ICF 230 3RD Ave, Suite 300 Waltham, NY 02451 ATTN: Kevin Palaia, P.G.	2026-04-08	QU-0227 **Reference** US CO ICF Brighton

Understanding of Project

This project, conducted under the EPA Office of Brownfields and Land Revitalization Technical Assistance framework, involves evaluating the feasibility of a geothermal heating and cooling system to support a proposed affordable housing redevelopment at 4640 Brighton Boulevard in Denver, Colorado. The Technical Assistance recipient, the Globeville Elyria Swansea Coalition, has secured funding from the Colorado Housing and Finance Authority and is advancing plans for a mixed-use development consisting of approximately 64 residential units, with a mix of two-, three-, and four-bedroom units, along with potential commercial space. The total gross floor area is estimated at 100,340 square feet. As the project design continues to evolve, this effort will rely on preliminary plans and assumptions to characterize building energy demand and site constraints.

The primary objective of this feasibility study is to assess the technical and economic viability of a geothermal heating and cooling system and to quantify potential greenhouse gas reductions relative to conventional HVAC systems. The analysis will include high-level building load estimation, site suitability screening, conceptual system design, and Class 4 cost and lifecycle analyses. The study will provide decision-support information to the Globeville Elyria Swansea Coalition and project stakeholders, coordinated through a kickoff and scoping process, to evaluate whether geothermal integration aligns with project affordability, site constraints, and broader sustainability and environmental justice goals. GEOptimize has been engaged by ICF, a subcontractor to Abt Global, to support this effort.

Scope of Work

Feasibility

- Develop a prototypical hourly energy model of the proposed group of buildings.
    - The model may include several iterations of the proposed buildings to determine the ventilation strategy, glass specifications and other potential energy efficiency measures that will result in the most cost-effective ground heat exchanger for the facility.
- Ground Loop Design (GLD) model to size the ground heat exchanger (GHX).

- Review of the geology of the site to determine the thermal properties of the soil and the most cost-effective drilling depth for potential drilling contractors in the area.
- Provide size and cost estimates for up to two GHX options to support a life-cycle cost analysis (LCCA), including operational costs and a financial evaluation of ROI, NPV, and simple payback.
    - Class 4 costing accuracy.
- Greenhouse gas emission analyses comparing the geothermal system with a baseline conventional HVAC system.

Information Required

- Site plan indicating square footages of various building types.
- Site plan indicating land area available for construction of the GHX.
- Geotechnical report.

Deliverables

- Preliminary GHX options and size/cost for feasibility analysis in a report.

Pricing

Description	Engineer 2 Hour's (200$/hr)	Supervisor Hour's (250$/hr)	Line Total
Building Energy Modelling	12	2	$2,900.00
GHX Modelling	32		$6,400.00
Class 4 Costing & LCCA	24		$4,800.00
Reporting & Meetings	20	4	$5,000.00
Total (USD)			**$19,100.00**

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